

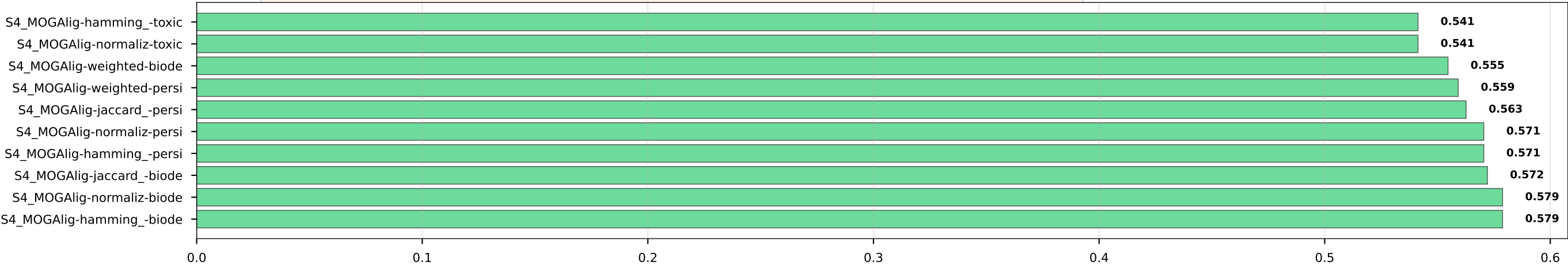
COMPREHENSIVE UBP STUDY - KEY RESULTS

Dataset: 1200 compounds across 23 chemical categories  
Strategies: 6 advanced UBP-based mapping approaches  
Metrics: 6 binary similarity metrics (Jaccard OffBits, Jaccard OnBits, Hamming variants)  
Properties: 3 chemical properties (Persistence, Biodegradability, Toxicity)

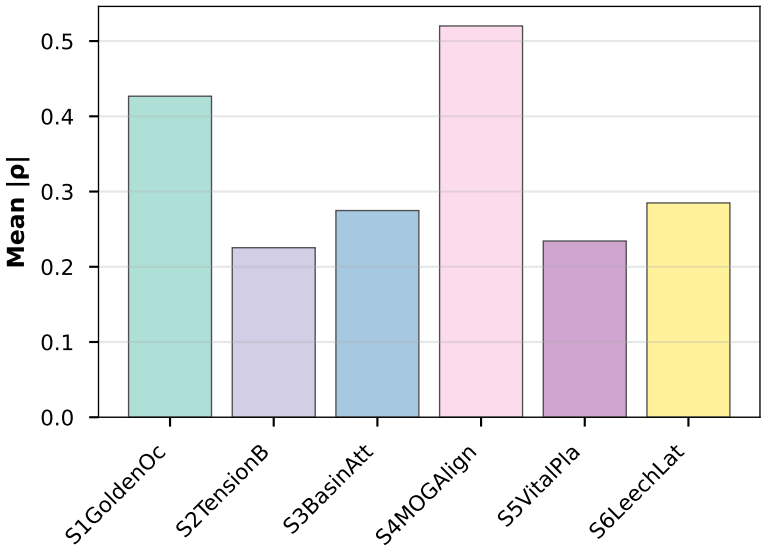
BEST CORRELATION:  $\rho = 0.579$   
Strategy: Strategy4\_MOGAligned  
Metric: Hamming Distance  
Property: Biodegradability

STATISTICAL SIGNIFICANCE:  
Total tests: 108  
Significant (raw  $p < 0.05$ ): 108 (100.0%)  
Significant (FDR  $< 0.05$ ): 108 (100.0%)

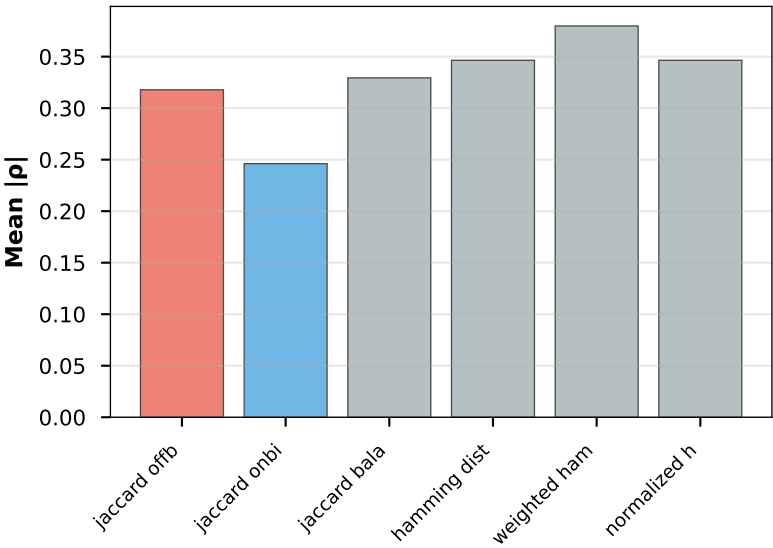
Top 10 Correlations



Strategy Performance



Spearman Correlation ( $\rho$ )  
Metric Performance



OffBits vs OnBits

